

Scientific Creativity as Constrained Stochastic Behavior: The Integration of Product, Person, and Process Perspectives

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Psychologists have primarily investigated scientific creativity from 2 contrasting in vitro perspectives: correlational studies of the creative person and experimental studies of the creative process. Here the same phenomenon is scrutinized using a 3rd, in vivo perspective, namely, the actual creative products that emerge from individual scientific careers and communities of creative scientists. This behavioral analysis supports the inference that scientific creativity constitutes a form of constrained stochastic behavior. That is, it can be accurately modeled as a quasi-random combinatorial process. Key findings from both correlational and experimental research traditions corroborate this conclusion. The author closes the article by arguing that all 3 perspectives—regarding the product, person, and process—must be integrated into a unified view of scientific creativity.

What enables scientists to make notable contributions to their disciplines? Why are some scientists more likely to make major discoveries than are others? These questions have obvious practical value and theoretical interest. Yet these questions have also provoked a wealth of contradictory answers. At least since the work of Francis Bacon and René Descartes, philosophers of science have tried to provide a logical basis for scientific discovery. The logic may be inductive or deductive, or some combination of the two, but the application should be deliberate and straightforward. Other thinkers have questioned whether scientific creativity can be subsumed under some totally rational method. These opponents include both philosophers, like Karl Popper (1959), and Nobel laureate scientists. For example, Max Planck (1949) claimed that creative scientists “must have a vivid intuitive imagination, for new ideas are not generated by deduction, but by an artistically creative imagination” (p. 109), and Albert Einstein reported that “To these elementary laws there leads no logical path, but only intuition, supported by being sympathetically in touch with experience” (as cited in Holton, 1971–1972, p. 97).

This philosophical debate has a parallel in research on the psychology of science (Feist & Gorman, 1998; Simonton, 1988b). On the one side are those psychologists who ground the creative process in the intellectual and dispositional capacity to generate remote associations, divergent thinking, rich imagery, and primary process (Guilford, 1967; Mednick, 1962; Rothenberg, 1987; Simonton, 1997; Suler, 1980), a position that dates as far back as Alexander Bain (1855/1977), William James (1880), and Ernst Mach (1896). On the other side of the debate are those psychologists who view creativity as a straightforward form of problem solving, a cognitive process that entails fundamentally logical mechanisms (Klahr & Simon, 1999; Newell, Shaw, & Simon, 1958). Perhaps the most conspicuous proponent of this view was

the Nobel laureate Herbert Simon (1973), who argued for an affirmative response to the question, “Does scientific discovery have a logic?”

Apparently, psychologists have come no closer to resolving this issue than have the philosophers. One possible reason for the lack of resolution is that the two camps represent diametrically opposed approaches to research—what Cronbach (1957) styled the “two disciplines of scientific psychology.” On the one hand, those advocating the rationalist position tend to favor experiments that concentrate on problem solving in laboratory settings (Hayes, 1989a; Klahr, 2000; Newell & Simon, 1972; Weisberg, 1988). Intimately tied to these endeavors are computer simulations—especially the so-called discovery programs—that implement the cognitive processes identified in the laboratory experiments (Langley, Simon, Bradshaw, & Zythow, 1987; Shrager & Langley, 1990). On the other hand, psychologists who see scientific discovery as entailing something more than mere logic tend to conduct correlational studies (Cattell, 1963; Eiduson, 1962; Eysenck, 1995; Roe, 1953). These studies use psychometric measures of cognitive style and personality traits, perhaps supplemented by biographical inventories that gauge familial and educational experiences. The goal in the latter tradition is to identify the distinctive psychological characteristics that separate highly influential scientists from their less successful colleagues.

Because the methods and aims are so disparate, the two research traditions find it difficult to agree on a common theoretical language to address the phenomenon of scientific creativity. So the data generated by one tradition are often taken as irrelevant to the data generated by the contrary tradition. Thus, Mansfield and Busse’s (1981) *The Psychology of Creativity and Discovery*, which represents the correlational tradition, virtually ignored the vast experimental literature on the subject. For example, the classic work of H. A. Simon (1977) receives not even a passing reference. Likewise, Eysenck’s (1995) book on creative genius contains only one short statement on the experimental tradition—a dismissal of attempts to model scientific discovery using computer simulations. On the other side of the disciplinary divide, Klahr’s (2000) *Exploring Science* discusses experimental research to the exclusion

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of correlational studies. Indeed, Klahr's list of the five general approaches to the empirical study of science altogether excludes personality and differential research.¹ Weisberg (1992), another proponent of the experimental perspective, went so far as to label the "creative personality" as one of the "myths of genius."

Notwithstanding these examples, each of these two opposing traditions may have captured a portion of the truth. Both perspectives may be required to comprehend scientific creativity in its full complexity. However, this necessity may become apparent only when this phenomenon is scrutinized from a totally different perspective. In particular, before discussing either psychometric assessments of scientists or laboratory experiments on scientific problem solving, I must first scrutinize what creative scientists actually do when they make bona fide discoveries. In other words, the information to be introduced is *in vivo* (naturalistic and behavioral) rather than *in vitro* (psychometric or experimental; cf. Dunbar & Blanchette, 2001). I argue that these real-life data all converge on a single implication: Genuine instances of scientific creativity at the behavioral level display the telltale characteristics of a *constrained stochastic process*. That is, creative behavior in science demands the intrusion of a restricted amount of chance, randomness, or unpredictability. After presenting the evidence on behalf of this proposition, I show that both correlational and experimental traditions provide a foundation for understanding the psychological origins of this stochastic behavior. The article ends by arguing that the two disciplines of psychology must be integrated with the behavioral data to obtain a complete understanding of scientific creativity.

Creative Behavior in Real-World Science

From time to time, scientists describe the mental processes that led to their discoveries (Hadamard, 1945; Helmholtz, 1898). Frequently, these descriptions imply cognitive operations that must be fundamentally stochastic in nature (Campbell, 1960; Simonton, 1999b). Probably the best known example is the introspective report published by the French mathematician and physicist Henri Poincaré (1921). In describing one discovery episode, he observed, "Ideas rose in crowds; I felt them collide until pairs interlocked, so to speak, making a stable combination" (p. 387). Poincaré (1921) compared these colliding images to "the hooked atoms of Epicurus" that jiggle and bump "like the molecules of gas in the kinematic theory of gases" so that "their mutual impacts may produce new combinations" (p. 393). Although this quote suggests a fairly random combinatorial process, it is not completely so. In particular, Poincaré (1921) claimed that the ideas participating in the chaotic collisions are provided during the preparation period of the creative process:

The mobilized atoms are . . . not any atoms whatsoever; they are those from which we might reasonably expect the desired solution. Then the mobilized atoms undergo impacts which make them enter into combinations among themselves or with other atoms at rest which they struck against in their course. (p. 389)

As a consequence,

the only combinations that have a chance of forming are those where at least one of the elements is one of those atoms freely chosen by our

will. Now, it is evidently among these that is found what I called the good combination. (Poincaré, 1921, p. 389)

The last sentence also suggests that only a tiny portion of the ideas generated by the chaotic mental process past muster by some scientific criterion. As Poincaré (1921) expressed it, "among the great numbers of combinations blindly formed . . . almost all are without interest and without utility" (p. 392). Hence, the probability of producing a "good combination" is very small.

Clearly this introspective report cannot be considered empirical proof that scientific creativity operates according to a stochastic mechanism. Such impressionistic information is fraught with potential biases and errors (Nisbett & Wilson, 1977; Perkins, 1981). Nevertheless, Poincaré's (1921) observations provide the foundation for an explanatory account of key phenomena associated with (a) scientific careers and (b) scientific communities.

Scientific Careers

From the standpoint of creativity, the most significant aspect of scientific careers concerns the output of products that have an impact on the discipline. After first reviewing the distinctive features of this phenomenon, I then provide a theoretical explanation in terms of a stochastic combinatorial mechanism. I conclude this section with a substantive discussion that consolidates the argument.

Creative Productivity

Scientists vary greatly in the influence they exert over their respective disciplines (S. Cole & J. R. Cole, 1973). One common means to distinguish the comparative impact is to consider the various honors and awards that a scientist can list on his or her curriculum vitae. At the highest level are those who have received international recognition, such as a Nobel Prize (Berry, 1981; Zuckerman, 1977) or at least election to scientific academies or societies in countries other than the nation in which they work (Candolle, 1873). At a bit lower grade are those who have earned national recognition, such as membership in the National Academy of Sciences (Feist, 1993; Over, 1981) or the Royal Society of London (Galton, 1874; Hudson, 1958). Lower still are those who are recognized by discipline-specific honors. An instance within psychology is the Distinguished Scientific Contribution Award of the American Psychological Association (Simonton, 1985). Below this is recognition like fellow status within a scientific society. Finally, at the bottom would be those active scientists who have not earned any special recognition for their work.

This conspicuous variation in recognition has two crucial features. First, individual differences in the scope or prestige in honors correlates highly with alternative indicators of distinction, including expert ratings or eminence gauged by biographical dic-

¹ The only notable personality study cited by Klahr (2000) is Roe's (1953; misspelled as Rowe in Klahr's work) examination of 64 eminent scientists. Even then, he only briefly mentions her discussion of thought processes, completely omitting her extensive analysis of motivational and personality attributes. Despite her prominence in the person-oriented literature in the psychology of science, she is a virtual unknown in the process-oriented literature (see also Boden, 1991).

tionaries, encyclopedias, or histories (Simonton, 1984d, 2002). In fact, alternative indicators of differential impact are demonstrably a function of a single latent variable (Simonton, 1991b). They thus all represent the same underlying construct, namely, the differential impact that a scientist has on his or her discipline. Second, and even more critical, this cross-sectional variation in disciplinary impact is ultimately grounded in individual differences in creative productivity (S. Cole & J. R. Cole, 1967; Feist, 1997; Simonton, 1991a). For instance, the total lifetime output of a nineteenth-century scientist predicts the probability that he or she will have an entry in a twentieth-century edition of the *Encyclopaedia Britannica* (Dennis, 1954a; Simonton, 1984b). Similarly, future Nobel laureates can be predicted on the basis of the total number of citations that scientists receive to their body of work (Ashton & Oppenheim, 1978), and yet the single best predictor of citations is the total number of publications (S. Cole & J. R. Cole, 1973; Simonton, 2002). Therefore, to comprehend how scientists vary in overall impact, it is essential to understand precisely how scientists differ in output.

This understanding must begin with the distinctive cross-sectional distribution of lifetime productivity (Simonton, 1997). For any given scientific discipline it is typical for the top 10% of scientists in total output to account for approximately half of the total contributions (Dennis, 1954a, 1954b, 1955). The dominance of this prolific elite appears because the productivity distribution is positively skewed, so that the modal level of lifetime output is a single contribution, ignoring those who published nothing at all. In more formal terms, rather than the normal curve most frequently observed for human traits, creative productivity is best described by a lognormal or exponential distribution (Huber, 1999; Lotka, 1926; Price, 1963; Shockley, 1957; H. A. Simon, 1955). Although this conclusion is based on total quantity regardless of the quality of the output, the same generalization holds for the production of high-impact work (Simonton, 1999b). For example, a small proportion of the scientists in any field accounts for a disproportionate number of citations in the research literature (Helmreich, Spence, Beane, Lucker, & Matthews, 1980; Myers, 1970; Redner, 1998; White & White, 1978). Yet it is significant that those who publish the most highly cited works also publish the most ignored works, so that quality is a probabilistic consequence of quantity (Simonton, 1997). In fact, the ratio of high-impact publications to total output—the “hit” rate—is uncorrelated with total output (R. A. Davis, 1987; Platz, 1965; Platz & Blakelock, 1960; Simonton, 1985; White & White, 1978). This has been called the *equal-odds rule* (Simonton, 1997; cf. Simonton, 1988a).²

Naturally, this principle operates only on the average. Some scientists are seeming perfectionists whose hit rates are higher than average, whereas other scientists are apparent mass producers whose hit rates are lower than average (J. Cole & S. Cole, 1972; Feist, 1997). Even so, these departures from statistical expectation are in line with the probabilistic nature of the association (Simonton, 1997, 1999b). Because the quantity–quality correlation seldom exceeds .70, and often falls closer to .50, the amount of variance shared between the two distributions does not exceed one half and frequently can be as low as .25. Ample latitude accordingly remains for the existence of considerable scatter around the regression line. Yet there is no evidence that these residuals or errors represent distinct types of scientists in a taxometric sense (Simonton, 1999b). On the contrary, the departures from statistical

expectation are precisely what would be anticipated from a statistical association of this magnitude. For example, suppose that the correlation between total publications and the number of highly cited publications is around .50. If the measures of quantity and quality are dichotomized at their respective medians, this correlation implies that 25% of scientists would be either high-quality and low-quantity perfectionists or high-quantity and low-quality mass producers. Even so, these exceptions to the rule merely confirm the statistical nature of the association. The scientists conforming to statistical expectation still outnumber the exceptions by three to one (J. Cole & S. Cole, 1972; Feist, 1997; Helmreich, Spence, & Thorbecke, 1981).

Complementary results emerge when the inquiry turns to longitudinal fluctuations in output within careers rather than cross-sectional distributions of output across careers (Simonton, 1988a, 1997). In general, the periods during the course of a scientist's career in which the most total output is produced tend to be those periods in which the most high-impact work appears (S. Cole, 1979; Over, 1988, 1989; Simonton, 1984b, 1985). In other words, quality is once again a probabilistic consequence of quantity, in line with the equal-odds rule (Simonton, 1988a, 1997). Stated more emphatically, the ratio of high-impact work to total output in any given career period neither increases nor decreases, nor does it assume some nonlinear form. Just as significant is the finding that the output of products tends to be randomly distributed across the career course (Huber, 1998a, 1998b; Huber & Wagner-Döbler, 2001). In particular, (a) there are no significant “runs” at any point in the career, whether at the beginning, middle, or end; and (b) the number of products appearing in any given time unit is Poisson distributed.³ These results hold for all alternative definitions of what constitutes a scientific product, such as patents awarded for new inventions or original articles published in refereed journals.

It is worth pointing out that the foregoing results apply not just to scientific creativity, but to any form of creativity (Huber, 2000; Quételet, 1835/1968; Simonton, 1977, 1997). In either case, quality is a probabilistic consequence of quantity, and the time pattern of output is random and Poisson distributed. For instance, the

² Suppose that, contrary to the equal-odds rule, the hit rate was a positive linear function of total output. If total lifetime output is given by T and the total count of high-impact works is given by H (where $T \gg H$), then the hit rate is $H/T = cT$, where c is a proportionality constant ($c > 0$). It necessarily follows, then, that $H = cT^2$. In other words, quality would be a *quadratic* rather than a linear function of quantity. Similarly, if the hit rate were a negative function of total output (i.e., $c < 0$), the function would remain quadratic, only with a decelerating rather than accelerating curve. Neither of these predictions is inconsistent with the empirical evidence gathered to date.

³ Like the normal distribution, the Poisson distribution can be derived from the binomial distribution with the parameters p (the probability of an event) and n (the number of trials). However, unlike the normal distribution, the derivation of the Poisson assumes that p is extremely small and n is extremely large. One final distinction is that unlike the equation for the normal distribution, which has two parameters (the mean μ and the variance σ^2), the equation describing the Poisson has only one because the mean equals the variance (i.e., $\mu = \sigma^2$, where $\mu = np$ as $n \rightarrow \infty$). In particular, the probability of i events occurring within a given unit is given by $P(i) = \mu^i e^{-\mu} / i!$ (where e is the exponential constant 2.718 . . . and $i! = 1 \times 2 \times 3 \times \dots \times i$).

output of compositions in classical music displays the same probabilistic pattern as the output of scientific publications or technological patents. Hence, in this respect, creativity must operate according to the same generic stochastic process in both the arts and the sciences.

Stochastic Explanation

The foregoing empirical findings can be explicated under the assumption that scientific creativity involves a stochastic process. To show how, I begin by making the following four assumptions:

1. Each scientific or technological specialty contains two essential components, namely what Csikszentmihalyi (1990, 1999) called the *domain* and the *field*. The domain consists of a large but finite set of facts, concepts, techniques, heuristics, themes, questions, goals, and criteria. These can be collectively referred to as the population of *ideas* that make up a given domain. These ideas are equivalent to the atoms in Poincaré's (1921) imagery. The field consists of all those individuals who are working with the set of ideas that define the domain. These are the scientists who are most likely to study, evaluate, and build on the discoveries and inventions created by others in the same specialty. It is they who ultimately decide the impact of any colleague's contribution (e.g., by their decisions regarding what to cite in their own publications).
2. Each scientist during the course of disciplinary education and training acquires his or her "sample" from the larger set of ideas. In other words, each individual who defines a field contains a unique subset of ideas that define the domain. Although a large portion of each creator's ideational sample is shared with one or more creators working in the same field, another portion of that sample is unique to each creator, owing to discrepancies in background and training. For instance, the scientist or inventor might have switched disciplines, bringing ideas that are alien to the new domain of choice (Hudson & Jacot, 1986; Simonton, 1984c).
3. The size of the ideational sample also varies across the scientists making up any given domain. That is, some individuals possess a large supply of disciplinary facts, concepts, goals, techniques, themes, heuristics, questions, and criteria, whereas other individuals possess a relatively diminutive sample of such ideas. For purposes of discussion, it can be assumed that the distribution of these sample sizes is roughly described by the normal bell shaped curve. This assumption merely reflects the likelihood that the psychological factors underlying the acquisition of this knowledge base are normally distributed (cf. Galton, 1869; Simonton, 1999c).
4. The disciplinary ideas making up each creator's sample are then subjected to free, relatively unconstrained, or quasi-random recombination, with the aim of finding original and useful permutations (Campbell, 1960; James, 1880). Because these combinations are "blindly

formed," to use Poincaré's (1921, p. 392) phrase, the scientist cannot accurately foresee which ideas will bear fruit and which will prove sterile (Simonton, 1995). As a consequence, these ideational variations actually have a relatively low probability of arriving at a creative combination, because a large amount of time is spent shifting through useless combinations. Fortunately, however, the low likelihood of success is partially compensated by the large number of trials. Each creator working within a given field presumably engages in this combinatory process for many years.

The above four assumptions are sufficient to explain the central features of creative productivity across and within scientific careers. In the first place, scientists differ in the total number of ideational combinations they can generate, depending on the size of the sample with which they work. However, although it was postulated that the sizes of these samples were distributed according to the normal curve, the distribution of the total output of ideas would not be (Simonton, 1988b, 1997). On the contrary, because the total number of potential combinations increases roughly exponentially with the number of available facts, concepts, goals, criteria, techniques, themes, heuristics, and questions entering the combinatorial hopper, the distribution of lifetime productivity should be closely approximated by a lognormal distribution (Simonton, 1988a). As expected, this distribution is positively skewed, with a long upper tail, so that a small minority of scientists accounts for a lion's share of the discipline's creative ideas. Even more important is the theoretical basis for the equal-odds rule. If the combinatorial process is truly blind, then what Poincaré (1921) styled "good combinations" will be randomly distributed across and within careers (Simonton, 1988b, 1997). Thus, with respect to cross-sectional data, hit rates should be uncorrelated with total output. Those scientists who produce the most ideas deemed worthy of attention by others in their fields would also tend to generate the most ideas that are ignored or even criticized by their colleagues. At least this association between quantity and quality would hold on average because by chance alone some scientists will have higher hit rates than average (*perfectionists*) and others will have lower hit rates than average (*mass producers*). Finally, if the combinatorial procedure is truly blind, then the equal-odds rule should hold within careers as well (Simonton, 1999b). Those periods in a scientist's career in which he or she generates the most combinations will be the periods in which the good combinations will most likely appear. Because the process exhibits no foresight, the good combinations that support high-impact products will be randomly distributed across the scientist's career. In addition, because the odds of finding a good combination are so small, the number of hits in any given time period should be described by the Poisson distribution. The Poisson distribution emerges whenever the probability of a hit is very small but the number of trials is very high (Haight, 1967; Molina, 1942). Hence, the central features of productivity across and within careers can be explicated by assuming that creativity operates like a stochastic combinatorial procedure.

Substantive Discussion

With some additional assumptions, it is possible to derive precise and testable predictions about how creative productivity varies

across and within careers. For example, by assuming that the stochastic combinatorial process consists of two steps (ideation and elaboration), one can generate predicted career trajectories for scientists who vary in creative output or who work in different disciplines (Simonton, 1984a, 1991a, 1997). Even more important, perhaps, are the connections between the hypothesized process and two critical features of the scientific enterprise: (a) the dynamic structure of research programs and (b) the low reliability and validity of peer review.

Dynamic structure of research programs. One phenomenon that is frequently evoked to illustrate the role of chance in scientific discovery is serendipity (Austin, 1978; Cannon, 1940; Mach, 1896; Roberts, 1989). This occurs when a scientist chances upon some discovery unexpectedly or by accident and without even intending to make the discovery in the first place. Serendipitous events often are responsible for totally unanticipated breakthroughs, such as the discovery of radioactivity, X-rays, and penicillin (Kantorovich & Ne'eman, 1989). Although serendipitous events are apparently exceptional, it is a commonplace to quote Louis Pasteur's comment that "chance favours only the prepared mind" (as cited in Beveridge, 1957, p. 46). That is, some scientists are luckier than other scientists. What is the basis for individual differences in luckiness?

The answer to this question may be found in the distinctive dynamic structure of successful research programs (Simonton, 1999b). This temporal pattern is inferred from the systematic analysis of laboratory notebooks (Gruber, 1974; Tweney, 1990), research bibliographies (Garvey & Tomita, 1972; Root-Bernstein, Bernstein, & Garnier, 1993), tape recordings of laboratory meetings (Dunbar, 1995, 1997), and responses to survey questionnaires (Hadamard, 1945; Hargens, 1978; R. J. Simon, 1974; Taylor, Locke, Lee, & Gist, 1984). Findings from these diverse sources all converge on the same set of basic conclusions. First, highly creative scientists seldom work on one project at a time. Instead, they tend to pursue several independent inquiries simultaneously. Moreover, each separate project varies tremendously regarding (a) the likelihood of its successful completion (i.e., high-risk vs. low-risk projects), (b) the intrinsic importance of the questions it addresses (e.g., breakthrough vs. mundane), (c) the relevance of that project to other projects at the time (i.e., peripheral vs. core), (d) the amount of progress that has so far been made in bringing the project to conclusion (e.g., initial pilot studies vs. writing up the final results), (e) the specific type of research (e.g., empirical, theoretical, literature review), and (f) the degree of concentrated effort the investigator is devoting to the project at any given time (e.g., high-priority vs. back burner inquiries). Finally, the status of these simultaneous endeavors will change dramatically over time. A project that might first attract the scientist's primary efforts may encounter some obstacle to further progress and thus be placed on hold while another project occupies the foreground. Because the scientist has several projects going on at once, cross-talk can take place among the projects (Miller, 2000). This cross-talk can often result in unexpected solutions. An event like this was illustrated by Poincaré (1921) in an introspective report:

I turned my attention to the study of some arithmetical questions apparently without much success and without a suspicion of any connection with my preceding researches. Disgusted with my failure, I went to spend a few days at the seaside, and thought of something

else. One morning, walking on the bluff, the idea came to me . . . that the arithmetic transformations of indeterminate ternary quadratic forms were identical with those of non-Euclidean geometry. (p. 388)

Because the research programs consist of a dynamic mix of different projects, with unforeseen and unforeseeable cross-talk among the various projects, distinguished scientists are necessarily exposed to serendipitous events. It should also be evident that under these conditions, the output of these scientists should have the appearance of being generated by a stochastic process. For instance, because of the chaotic nature of the process, high-impact discoveries should be randomly distributed over the career course. As indicated in the Poincaré (1921) passage just quoted, sometimes the solution to an important problem must await the resolution of some other substantive question that is not even deemed relevant at the time.

Low reliability and validity of peer review. The preceding discussion greatly underestimates the chaotic nature of scientific research programs. There are many other external events that impinge on scientists that can provide the impetus for serendipitous directions in the course of inquiry. For instance, notable scientists tend to read widely, including in areas outside their main discipline (Dennis & Girden, 1954; R. J. Simon, 1974). This reading habit increases the odds that they might chance upon discoveries that turn out to be surprisingly relevant to one of their current projects. A classic example is how Charles Darwin's causal reading of the *Essay on the Principle of Population* by Thomas Malthus inspired the crucial concept of the struggle for existence (Darwin, 1892/1958). In addition, highly visible scientists are frequently asked to evaluate grant proposals submitted to funding agencies and manuscripts submitted to refereed journals. The ideas contained in these submissions may unexpectedly activate new trains of thought in their own research. This stimulation is probably especially potent for those submitted proposals and manuscripts that concern research outside the reviewer's core research area.

Although such participation in peer review would clearly provide more or less random inputs into the scientists serving as reviewers, the practice of peer review might seem to undermine the validity of the equal-odds rule. When proposals or manuscripts are sent out for evaluations, the reviewers are usually asked to apply an explicit set of criteria (Gottfredson, 1978; Wolff, 1973). These criteria collectively determine the quality of the submission, presumably according to scientific or objective standards (Gottfredson, 1978). Thus, the very existence of these criteria implies that scientists can readily discriminate good from bad research (Sternberg & Gordeeva, 1996; Wolff, 1973). If so, then why should quality be a probabilistic consequence of quantity? From the very start of the career, and throughout its course, each scientist receives constant feedback from colleagues about the quality of his or her work. Hence, over time, each scientist should internalize those criteria and eventually be able to augment the ratio of hits to total attempts. Not only would the quality ratio increase over the career course, but also the most prolific scientists—those who have passed through the peer review cycle most frequently—should have higher hit rates than their less productive colleagues.

This apparent paradox can be resolved without compromising the thesis that scientific creativity entails a stochastic mechanism.

It turns out that peer review is too unreliable to provide useful information about the quality of a scientist's research efforts. It is not a matter that scientists do not agree on the criteria by which research must be judged, for on that score there exists a respectably strong consensus (Sternberg & Gordeeva, 1996; Wolff, 1970). Instead, the problem arises in the application of the criteria. In the first place, referees agree very little in their evaluations of grant proposals and submitted manuscripts (Cicchetti, 1991; S. Cole, 1983; S. Cole, Cole, & Simon, 1981; Lindsey, 1988; Scott, 1974). In fact, the degree of agreement is only slightly higher than what would be obtained if the assessments were generated by a table of random numbers (Lindsey, 1988). Even worse, the criteria have virtually no predictive validity (S. Cole, 1983; Gottfredson, 1978; Shadish, 1989). Grant proposals that receive the highest ratings do not necessarily generate high-impact research, nor do the most highly rated manuscripts earn the most citations in the journal literature. In short, the reliability and validity of peer review are insufficiently strong to provide meaningful feedback on the quality of a researcher's work.

This conclusion may seem a bit surprising, but it makes more sense after some additional scrutiny. Many of the criteria that reviewers are asked to apply require judgments that are extremely difficult if not impossible to make. For instance, to predict the likely long-term impact of a given scientific inquiry would demand that the evaluator be some kind of prophet who can anticipate the future direction of the field. On the other hand, the criteria that are most easily applied are likely to be the least useful in making assessments. A good example is the evaluation of a study's methodology. Because most scientists acquire considerable methodological expertise during graduate and postdoctoral training, it is relatively rare for a grant proposal or submitted manuscript to have fatal flaws by this criterion. Given this ceiling effect, the evaluation's center of gravity must shift to more subjective judgments, such as the expected long-term impact of the investigation. Even worse, an objective criterion like method is often compromised by the more subjective criteria (Wilson, DePaulo, Mook, & Klaaren, 1993). In particular, methodological inadequacies are more likely to be overlooked in scientific inquiries that treat questions that are judged to be substantively significant.

All told, the evaluations that investigators receive from peer review have minimal informational value. Indeed, if the feedback is practically random, it could serve only to enhance the stochastic nature of the creative process. The evaluations add another source of chance to the serendipitous influx of other factors, such as the cross-talk among simultaneous projects and the unanticipated impact of outside reading.

Scientific Communities

Psychologists often view scientific creativity as a process that takes place within the individual. Yet as the operation of peer review suggests, the individual scientist necessarily works within the context of a larger community of scientists (Crane, 1972). This community includes those individuals who make up the scientist's field. As noted earlier, the field consists of that group of colleagues who define the substantive problems and methodological techniques that distinguish each scientist's domain of creative activity (Csikszentmihalyi, 1990, 1999). Moreover, once the analysis is

expanded to encompass many scientists working within the same discipline, a new phenomenon becomes possible: scientific multiples (Lamb & Easton, 1984; Merton, 1961b).

Multiple Discovery

A multiple discovery is said to take place whenever two or more scientists independently make the same contribution. The history of science is replete with conspicuous examples: the calculus originated by Newton and Leibnitz; the theory of evolution by natural selection proposed by Darwin and Wallace; the laws of genetic inheritance discovered by Mendel, De Vries, Correns, and Tschermak; the phenomenon of classical conditioning studied by Pavlov and Twitmyer; and the theory of emotion put forward by James and Lange. Sociologists and anthropologists have traditionally used the multiples phenomenon as evidence against the role of the individual creator in scientific discovery (Kroeber, 1917; Merton, 1961b; Ogburn & Thomas, 1922). Rather than being the product of scientific genius or creativity, each discovery is the inevitable outcome of sociocultural processes. Even so, it is possible to provide an interpretation of multiples that preserves the place of the individual scientist and even substitutes a stochastic mechanism for the putatively deterministic process (Simonton, 1987, 1999b). This alternative interpretation depends on recognizing that this phenomenon has four critical characteristics:

1. *Distribution of multiple grades.* Multiples vary in the number of rival claimants to the discovery or invention. If the number of independent claimants for any given multiple defines its *grade* (Simonton, 1979), then scholars have collected examples that range from Grade 2 to Grade 9 (Simonton, 1986d). However, it must be recognized that not all the various grades are equally probable. In general, the frequency distribution declines as the grade increases. Hence, doublets are the most common, triplets the next most common, and so on, with high-grade multiples represented by only one or two cases. For instance, the most comprehensive collection gathered to date claimed 449 doublets but only 104 triplets, 18 quadruplets, 7 quintuplets, and 1 octuplet (Simonton, 1979).

2. *Temporal separation of multiple discoveries.* A striking feature of many multiples is that two or more workers may offer the same contribution at almost the same time. In one study of 264 multiples, for example, fully 20% of the independent contributions took place within a 1-year interval (Merton, 1961b). However, it must be considered just as remarkable that many of the multiple participants are separated by many more years. In fact, according to the same inquiry, 34% of the multiples required at least a decade to transpire before the duplications ceased (Merton, 1961b). Indeed, occasionally hundreds of years will divide the first and last instance of a multiple (e.g., the discovery of the Eustachian tubes, the principles of hydrostatics, the transit of Venus across the solar disk). It is telling that the frequency distribution of time lags is very similar to that for multiple grades. The larger the magnitude of temporal separation, the lower its expected frequency (Simonton, 1987).

3. *Individual variation in multiple participation.* Some scientists are more likely to see their work anticipated by other colleagues or to find themselves entering into priority disputes (Hagstrom, 1974; Simonton, 1979). However, those who participate in many multiples tend to differ systematically from those who are

involved with few or none at all (Simonton, 1999b). First, the higher the number of total publications a scientist produces, the larger the number of multiples in which he or she participates (Simonton, 1979). Second, those scientists who are active in large fields with many active colleagues tend to get involved in multiples at higher rates than do so-called lone wolves who carve out a specialty uniquely their own (Hagstrom, 1974).

4. Degree of multiple identity. The lists of multiples on which some of the foregoing conclusions have been based have often been criticized. One of the most common complaints is that some supposed cases include discoveries or inventions that are actually far from identical (Patinkin, 1983; Schmookler, 1966). That is, the independent contributions can be lumped together only by imposing extremely inclusive, genetic categories that have no real meaning. For instance, the supposed multiple invention of the steam turbine places under one generic category machines that have dramatically different designs and that operate under totally different physical principles (Constant, 1978). To be sure, genuinely identical multiples do occur, such as the Pelton water wheels (Constant, 1978), but these tend to be relatively rare. In general, just as multiples vary regarding their grade and temporal separation, so can they vary according to the degree to which they actually represent similar contributions. This variation is evident in the interference proceedings that take place before the U.S. Patent Office when one patent application is claimed to infringe upon an earlier patent (Schmookler, 1966). Typically, the overlap involves only one feature out of over a hundred that may be specified in the application. It is extremely rare for two features to be involved, and genuine duplication practically never occurs.

Stochastic Explanation

Many of the empirical results just reviewed can be readily explicated using the principles already postulated in the earlier discussion of creative productivity. Most notably, not only may the occurrence of multiples be explained, but also the specific distribution of multiple grades (Simonton, 1999b). In the first place, the duplicate efforts merely reflect the fact that most disciplines consist of several scientists applying quasi-random combinatory processes to ideational samples that share a core of domain-specific concepts, facts, techniques, goals, criteria, heuristics, themes, and questions. By chance alone, then, two or more scientists are bound to generate the same ideational combination. Nevertheless, given the low probability of finding a good combination, the frequency of multiple grades should closely follow a Poisson distribution (Price, 1963; Simonton, 1978). Again, the latter distribution is characteristic of stochastic processes in which the probability of success is extremely low but the number of trials is extremely high. In fact, chi-square goodness-of-fit tests have demonstrated that the Poisson distribution accurately describes every frequency tabulation of multiple grades so far published (Simonton, 1978, 1979). Any departures from the frequencies predicted by the Poisson process can be attributed to sampling errors.

By the same token, it is readily understood why the most prolific scientists tend to participate in more multiples (Merton, 1961a; Simonton, 1979, 1999b). Just as those scientists who can generate more ideational combinations tend, on average, to produce more good combinations, so are they more likely to come up with combinations similar to those conceived by colleagues working

with samples containing roughly the same set of questions, facts, concepts, techniques, criteria, heuristics, themes, and goals. Still, it is manifest that absolutely identical duplicates would be rather rare. Just as two siblings from the same parents will almost never inherit the same set of genes, except when they come from the same fertilized egg (i.e., monozygotic twins), so will the discoveries or inventions constituting a putative multiple consist of a different mix of combinatorial components. Indeed, the distinctiveness of the supposed duplicates is all the more assured by the earlier assertion that scientists working within the same discipline seldom possess identical samples from the domain.

The other central characteristics of multiples can be explained by adding just two more assumptions:

1. Most obviously, the number of individuals making up a field varies considerably from one field to the next. Some disciplines or specialties attract many participants, whereas others are much less popular. This means that some disciplines have dozens, even hundreds, of scientists subjecting some of the same facts, concepts, techniques, heuristics, criteria, themes, questions, and goals to the free combinatory process, whereas other disciplines may have only a half dozen or so scientists with highly overlapping ideational samples.
2. When scientists come across a good combination, they will make others aware of the idea through conference presentations, journal publications, patent applications, or other vehicles of professional communication. The successful dissemination of a particular combination necessarily preempts others from arriving at that same combination (Merton, 1961b). Once creators in the same field find out that they have been "anticipated," they move to some other topic or issue, often incorporating the pre-empting discovery into the repertoire of ideas that will be subject to future combinatory processes. Yet this communication process need not be instantaneous. Publication lags of various kinds, coupled with barriers set up by linguistic, cultural, and perhaps disciplinary boundaries, would cause a certain amount of delay. Nonetheless, as time transpires, the innovation will become more widely disseminated, so that the probability of a duplicate effort will progressively shrink, eventually reaching zero. In addition, it can be safely assumed that the communication process has become more efficient over the course of scientific history. The advent of the printing press, professional journals, scientific conferences, and most recently, the Internet served to accelerate the diffusion of new ideas in science.

The first assumption is required to explain why scientists are most likely to participate in multiples if they are active in large fields with many colleagues subjecting similar samples of ideas to the same combinatorial process (Simonton, 1999b). By chance alone, the more scientists there are who are manipulating the same set of concepts, facts, techniques, questions, criteria, heuristics, themes, and goals the greater the odds that two or more scientists will find themselves anticipated. The second assumption is needed to explain why multiples vary in the temporal separation of their

duplicates. This assumption can be formally expressed in terms of the addition of a *contagion* component to the stochastic process that accurately predicts the empirical distribution of multiple grades (Brannigan & Wanner, 1983a; Simonton, 1986b, 1999b). The result is a stochastic model that also predicts the observed distribution of time lags between the first and last duplicate in a multiple (Brannigan & Wanner, 1983b; Simonton, 1986c). Furthermore, the parameters of this model show that the expected degree of temporal separation has indeed been declining (Brannigan & Wanner, 1983b). Current forms of scientific communication are so efficient that independent discoveries must be practically simultaneous for a multiple to occur at all.

Substantive Discussion

The stochastic explanation just presented manages to explicate all of the main features of multiple discovery without having to posit some mysterious form of sociocultural determinism. The causal role of the *zeitgeist*, or spirit of the times, has been reduced to the collection of facts, concepts, techniques, heuristics, questions, criteria, themes, and goals that enter the disciplinary pool from which creators draw their sample of ideas for their combinatory activities. This repository supplies the necessary conditions for all discoveries made in a specific period of a discipline's history. Hence, if a prerequisite idea is missing from the given *zeitgeist*, the corresponding ideational combination cannot emerge. For example, because a calculus cannot appear without an analytic geometry having first appeared, Newton's mathematical achievements presuppose those of Descartes. Ironically, this very necessity augments even more the probabilistic nature of scientific creativity. A scientist may try to solve a problem for which the prerequisite elements are not in place and thus be doomed to failure without any awareness that a piece of the puzzle is missing. Many astronomers attempted to explain a certain irregularity in Mercury's orbit without knowing that an explanation was impossible from the perspective of Newtonian mechanics. In a sense, the probability of a solution was zero until the advent of Einstein's general relativity theory.

But it must be stressed that this repository of domain-specific ideas provides the necessary but not sufficient basis for the combinations that lead to major discoveries (Simonton, 1986c, 1987). Just because the prerequisite ideas are in place for a particular discovery by no means guarantees that the discovery will be made right away, or even appear at all. As a consequence, many discoveries consist of ideas that have been in existence for decades, if not centuries, the delay reflecting the stochastic process by which the ideational combinations are generated. An example is paper chromatography, which earned Martin and Synge a Nobel Prize even though Martin himself admitted that "all the ideas are simple and had peoples' minds been directed that way the method would have flourished perhaps a century earlier" (Daintith, Mitchell, & Tootill, 1981, p. 531). The same inference of unpredictability can be drawn from rediscoveries in which one multiple is separated from the other by decades or more. Because Mendel's findings were independently rediscovered 35 years later, had he not carried out his experiments, genetics would have been discovered for the first time in 1900 rather than 1865.

Notwithstanding the fact that the disciplinary *zeitgeist* has been

placed in a relatively passive role, the repository of available ideas is more dynamic than first meets the eye. Each time a scientist successfully creates and communicates a new combination of these disciplinary ideas, that combination enters the ideational pool from which future discoveries or inventions can be made. Once Newton uses gravitational mechanics to explain the orbits of the known planets, subsequent scientists could use the same laws to predict the existence of previously unknown planets, such as Neptune. Furthermore, these additions to the repository vary greatly in their capacity to stimulate new scientific discoveries. For example, discoveries that are anomalous with respect to the scientific paradigms of the day can play a powerful role in scientific revolutions (Kuhn, 1970). Thus, research on blackbody radiation had this catalytic role in the advent of Planck's quantum theory, which led to the quantum revolution.

Finally, as the cumulative body of scientific knowledge increases, the number of potential ideational combinations expands dramatically, generating an information explosion in science. Expanding the possibilities all the more is the growth in the number of scientists who are engaged in the combinatorial process. The two upward trends in the number of ideas and the number of scientists taken together imply that the production of new discoveries should become highly accelerated. Not only is there ample evidence that this is the case (Fowler, 1987; Price, 1963), but also there is evidence that the creativity of individual scientists has become amplified (Price, 1963; Simonton, 1992). The larger the scientific community in which an individual is embedded, and the richer the cumulative body of scientific knowledge, the greater the creativity that can be displayed by the most prolific scientist in a given field. This effect is implicit in the Price law, which claims that if k represents the number of scientists active in a field, the square root of k represents the number of scientists who are responsible for half of all contributions to the discipline (Price, 1963; cf. Allison, Price, Griffith, Moravcsik, & Stewart, 1976). According to this law, as the number of scientists increases, a smaller proportion of scientists will account for half of the work. In a sense, the discipline becomes more and more elitist as the field grows in the number of active participants.

It is worth noting an interesting convergence between what has been learned about scientific creativity at the individual level and that occurring at the community level. In the earlier discussion on scientific careers, I observed that the number of discoveries in a given unit of time is described by a Poisson distribution, whereas in the current section on multiple discovery I mentioned that the frequencies of various multiple grades are also described by a Poisson distribution. These two Poisson distributions have very similar parameters (i.e., $\mu \leq 1$). The convergence not only confirms that Poincaré's (1921) good combinations constitute low-probability events but also implies that the estimated probability would be about the same for individuals and communities. At the level of the individual scientist, it takes many combinatorial trials to get one hit, whereas at the community level it takes many individuals recombining the same set of ideas to obtain a single success. Rarely does a scientist have more than one good idea in any given period, and just as rarely do two or more scientists within the same field come up the same good idea. The odds are comparably low in either case.

Corroborative Implications of the Two Psychologies

The statistical analyses that dominate psychology's journals tend to test the null hypothesis that the empirical results—whether they be mean differences or correlation coefficients—can be attributed to mere chance. By chance alone, it is possible to obtain sample means that look different and correlation coefficients that appear much larger than zero, even when in the general population the means are actually identical or the correlations are exactly zero. In an analogous fashion, the stochastic interpretations given above can be said to represent the null hypothesis that chance can account for all of the central empirical features of two key behavioral phenomena—scientific productivity and multiple discovery. In this case, however, chance is not sampling error but rather a hypothesized stochastic process. Beginning with the postulate that scientific creativity involves the quasi-random combination of the facts, concepts, techniques, heuristics, themes, questions, goals, and criteria that define a domain, a host of detailed features of scientific careers and communities can be explained and predicted. There is no need to hypothesize anything more mysterious, whether genius or *zeitgeist*, logic or inspiration.

Yet this stochastic conception of scientific creativity might appear unpalatable to psychologists working in either of the discipline's two main traditions. Those operating from the correlational perspective may believe that the role of creative genius has been overlooked. Instead of having exceptional abilities or extraordinary traits, great scientists appear to be the mere beneficiaries of blind luck. In contrast, psychologists aligned with the experimental perspective may think that the role of logic has been ignored. Any process that depends so heavily on chance appears to be inherently illogical. However, below I argue that the stochastic nature of actual creative behavior in the sciences can be derived directly from central results from both research traditions.

Correlational (Psychometric) Tradition: The Creative Person

The psychological study of distinguished scientists began with Galton's (1874) survey of Fellows of the Royal Society of London. Since then, a vast literature has accumulated on the characteristics of creative scientists (Feist, 1998; Mansfield & Busse, 1981). This body of knowledge forms part of a more comprehensive literature on the personal attributes of creative individuals of all kinds (Martindale, 1989; Simonton, 1999a). As mentioned earlier, in many respects creativity in science operates no differently than creativity in other domains. Yet in certain limited ways, creative scientists can be distinguished from their counterparts in other areas of creative achievement. Hence, I first treat creative individuals in general before turning to creative scientists in particular.⁴

Creative Individuals

If taken on face value, the psychometric literature on the creative person has come up with several findings that would suggest that such individuals have a distinctive capacity for engaging in a stochastic combinatorial process. This capacity is based on both cognition and personality.

Cognition. Many tests that purport to assess individual differences in creativity are rooted in the assumption that people vary in

their ability to engage in certain cognitive processes (Guilford, 1967). It is also apparent that the operation of these mental processes suggests that creativity must function in a stochastic manner (Simonton, 1999b). The best example is Mednick's (1962) well-known Remote Associates Test (RAT). This test was based on the premise that creativity involves the ability to make remote associations between separate ideas. Highly creative individuals are said to have a flat hierarchy of associations as compared with the steep hierarchy of associations of those with low creativity. The chancy consequences of such flat associative hierarchies is apparent in the following three considerations (for more details, see Eysenck, 1995; Simonton, 1999b):

1. A flat associative hierarchy means that for any given stimulus, the creative person has many associations available, all with roughly equal probabilities of retrieval. Persons who are very low in creativity, by comparison, have steep associative hierarchies in which any given stimulus elicits only one or two responses in a highly predictable fashion. Clearly, the thought processes of individuals with flat associative hierarchies should be more unpredictable than the more stereotyped thinking linked with steep hierarchies. Whereas the latter responds in the same way to the same stimulus time after time, the former responds more variably because the most probable response at any one moment is largely decided by the psychological equivalent of a coin flip.
2. Besides the almost equiprobable nature of the associations, flat associative hierarchies are necessarily more susceptible to relatively minor priming effects (Simonton, 1999b). An unexpected input of an external stimulus or an internal train of thought can easily change the relative probabilities of the associations making up the hierarchy. Hence, for highly creative persons, the most probable sequence of associations would be highly unstable over time, making their behavior more unpredictable relative to those less creative.
3. Not all associations making up a given hierarchy are equally likely to evoke that remote connection that provides the solution to a given problem. Rather, the vast majority of associations lead to cul-de-sacs during the incubation phase of the creative process. Given that successful associative linkages are so rare, the odds of chancing upon the most fruitful associative sequence become quite unpredictable. Sometimes the correct association arrives early, sometimes a bit later, and yet other times even later still. Essentially, the incubation period

⁴ Here I deliberately ignore individual differences in motivational variables. Although it is patent that highly creative individuals are extremely goal oriented and display a tremendous amount of drive, energy, and willful persistence and determination, these volitional features are not unique to outstanding creators. On the contrary, such motivational traits are associated with high achievement in a diversity of domains, including leadership, sports, performance, and entertainment (Cox, 1926; Galton, 1869; Simonton, 1994). Hence, the focus here is what actually distinguishes highly creative persons from those who are much less so.

behaves like a game of craps in which a winning throw has the same chance of appearing no matter how long the gambler has been playing the game.

These three points together imply that scientific creativity should operate very much like a Poisson process. That is, the flat associative hierarchy, in effect, will yield a large number of trials, each having a very low probability of producing a successful outcome.

Personality. Apart from the cognitive capacity for remote association, highly creative individuals feature dispositional traits that would render their thoughts and behavior less predictable and hence more stochastic (Simonton, 1999b). In the first place, creativity is positively associated with openness to experience (King, McKee Walker, & Broyles, 1996; McCrae, 1987), a disposition that includes a diversity of interests and hobbies, a preference of complexity and novelty, and a tolerance of ambiguity (Barron, 1963; G. A. Davis, 1975; Gough, 1979). A necessary repercussion of this characteristic is that the intellect of a highly creative individual is more frequently exposed to a tremendous variety of extraneous influences. Such diverse input can deflect the associative process toward more unanticipated directions. Indeed, such priming effects become all the more unpredictable when coupled with the flat associative hierarchies just mentioned (Simonton, 1999b). If a large number of associates have nearly equal probabilities, then experiential openness would destabilize the relative order of those associates from one moment to the next.

A second trait of highly creative individuals operates in a more complicated manner: The creative person has some inclinations toward psychopathology (Juda, 1949; Ludwig, 1995; Post, 1994). This propensity is revealed by higher than average scores on the clinical scales of the Minnesota Multiphasic Personality Inventory (Barron, 1969) as well as elevated scores on the Psychoticism Scale of the Eysenck Personality Questionnaire (Eysenck, 1993; Rushton, 1990). These empirical results may partly reflect the creator's need for independence or autonomy in order to avoid the constraints of conventional views. Yet another portion of these findings may be more directly connected to the cognitive process behind creativity. Eysenck (1994) has shown, for example, that psychoticism is positively correlated with the capacity for unusual associations and with appreciation for highly complex stimuli. Both of these outcomes would have the consequence of making the thought process more unpredictable and hence more stochastic. It is interesting that Eysenck (1995) has also shown that psychoticism is positively associated with a relative inability to filter out extraneous information (i.e., weakened latent inhibition and negative priming). Because of such attention failures, the creative mind is more likely to be bombarded by stimuli that will influence the outcome in unexpected ways (see also Wuthrich & Bates, 2001). Not surprisingly, reduced latent inhibition also has a strong association with openness to experience (Peterson & Carson, 2000) and to creativity assessed by both psychometric assessments and behavioral measures based on actual creative achievements (Carson, Peterson, & Higgins, in press).

In light of the above remarks, it is not easy to separate the cognitive and personality correlates of creativity. Indeed, the two sets of individual-difference variables may have a complex developmental etiology. For instance, if a person displays an exceptional openness to experience from early childhood, then it seems

likely that the person will reach adulthood with a richer and more diverse associative network (Simonton, 1999b). Even so, it is not necessary to settle on the details of their causal relation to appreciate that these intellectual and dispositional traits would necessarily operate to make the creative process more stochastic in basic operation.

Creative Scientists

Although research findings on the creative individual appear supportive of the hypothesized stochastic process, ample evidence shows that it takes more than just this cluster of traits to produce a creative scientist. Two considerations are especially crucial in this regard:

1. To possess the traits of the creative person is not tantamount to that individual having the capacity for creative achievement (Eysenck, 1995). On the contrary, creative productivity depends on the acquisition of domain-relevant expertise (Amabile, 1983; Hayes, 1989b). Thus, creative scientists must first acquire the facts, concepts, criteria, questions, techniques, heuristics, themes, and goals that define their chosen domain. It has been estimated that it usually requires at least a decade of extensive study and practice to attain world-class expertise in any domain of achievement (Hayes, 1989a; Ericsson, 1996a), and there is no reason to doubt that scientific creativity is any different (Ericsson, 1996b; Simonton, 2002). Hence, to some extent creative scientists make scientific discoveries rather than write poems or design buildings because they have mastered the materials of science rather than those of literature or architecture. Even so, domain expertise may be necessary for scientific creativity, but it is no more sufficient than are the individual-difference variables (Simonton, 1996). An expert can be distinguished from a creator in that the former has mastered a domain without having the ability to make novel contributions to the domain. In other words, experts have the knowledge and skill without the originality, whereas creators have knowledge and skill combined with the originality provided by their cognitive and dispositional constitution (see also Rostan, 1994).
2. Nevertheless, the personal characteristics of creative individuals are not identical no matter what the domain may be (Feist, 1998). Yet this outcome would not be expected according to theory. Although all forms of creativity require the intrusion of probabilistic processes, not every type of creativity is equally stochastic. That is, the various domains differ in the extent to which that stochastic process is constrained. These constraints include restrictions on (a) the ideas that enter the combinatorial process, (b) the degree to which the combinatorial process is fully random, and (c) the criteria that are applied to determine what constitutes a good combination (Simonton, 1999b). In general, scientific creativity is far more constrained by such standards than is artistic creativity. There are even differences within both the sciences and the arts regarding the magnitude of constraint.

In the arts, for example, formal and academic styles tend to be much more constrained than surrealist and expressionist styles (Ludwig, 1998). In addition, as an artistic style develops over time, the combinatorial process tends to become ever more free, and eventually even the constraints imposed by the artistic conventions tend to break down (Martindale, 1990). In the case of the sciences, scientists who practice what Kuhn (1970) termed *normal science* are engaged in “puzzle-solving” research that is highly constrained by a given paradigm. In contrast, scientists who revolutionize their disciplines by offering an entirely new paradigm operate under fewer constraints. In fact, a significant feature of a revolutionary paradigm is that it replaces the old restrictions with entirely new constraints on what counts as normal science (Kuhn, 1970). In any case, the individual-difference variables that are most clearly implicated in supporting the stochastic process should vary according to the extent that that process is constrained. For example, as compared with artistic creators, scientific creators should exhibit steeper associative hierarchies, less openness to experience, and lower incidence rates of psychopathology. Yet all of these characteristics should be higher in creative scientists than in their less creative colleagues. So far, the empirical literature endorses these theoretical expectations (e.g., Gough, 1976; Ludwig, 1992).

Most of the cognitive and dispositional characteristics required for creativity have substantial heritability coefficients (Eysenck, 1995; Simonton, 1999c). Therefore, to some as yet undetermined extent it is possible to speak of innate scientific talent (Bramwell, 1948; Brimhall, 1922, 1923a, 1923b; Galton, 1869, 1874). Thus, it may even be argued that genetic endowment may partially influence whether a scientist is disposed toward normal or revolutionary science. Even so, it is likely that most of the cross-sectional variation in scientific creativity can be ascribed to environmental factors, such as family background and educational achievement (Simonton, 1988b, 2002). This likelihood enables the derivation of several predictions about how developmental experiences might correlate with the type of creativity displayed (Simonton, 1999b). Specifically, individuals contributing to domains in which the stochastic combinatorial process is highly constrained will most likely emerge from highly conventional and stable family backgrounds and will exhibit high levels of educational attainment, as revealed by high grades and advanced degrees. Creators active in domains in which the process is much more unconstrained will most likely grow up in family environments that are far less conventional and stable, and their education will be less accomplished. There already exists ample evidence supporting these predictions (e.g., Raskin, 1936; Simonton, 1986a; Sulloway, 1996). For example, Nobel laureates in the sciences come from much more stable backgrounds than do laureates in literature (Berry, 1981), a contrast that reflects a parallel contrast in gifted and talented children (Schaefer & Anastasi, 1968). In a sense, the more unconstrained the stochastic process is required to be, the more unpredictable is the environment in which the creator appeared.

Experimental (Laboratory) Tradition: The Creative Process

Although the focus of the correlational tradition is the individual scientist, the experimental tradition concentrates on the creative process, especially in the context of problem solving. Much of this research is rooted in Newell and Simon's (1972) *Human Problem Solving*. According to this seminal work, a problem consists

of an initial state, a goal state, and a set of permissible transformations from one state to another (called “operators”) that, when executed in a correct sequence, result in a solution path from the initial state to the goal state, via a series of intermediate states and subgoals. Operators have constraints that must be satisfied before they can be applied. The set of states, operators, goals, and constraints is called a “problem space,” and the problem-solving process can be conceptualized as a search for a path that links the initial state to the goal state. (Klahr, 2000, p. 23)

The crux of problem solving is thus the discovery of the successful path. To make this discovery, the individual must first acquire a sufficient amount of domain-relevant knowledge and skill and thereby become an expert rather than a novice in a particular discipline. This acquired expertise usually includes specialized techniques that enable problems to be solved in a more or less algorithmic fashion. The expert needs only to identify the class of problems in which a given problem belongs and then apply the appropriate algorithm for its successful solution. These problem-solving techniques are so powerful that they have been termed *strong methods* (Klahr, 2000).

Thus far, it would appear that this view of the creative process is antithetical to the stochastic models put forward to explain creative behavior in real science. For instance, this conception seems inconsistent with the equal-odds rule. Once a scientist acquires the problem-solving techniques appropriate to his or her discipline, then solution should follow solution in an almost inevitable progression. If any longitudinal trend occurs at all, it should be pointed in the direction of increased success rates with enhanced experience. In addition, it seems reasonable to infer that those scientists who have produced the largest output should enjoy higher hit rates than those whose low output provided them with much less practice in the application of the disciplinary algorithms. Certainly nothing in the application of such strong methods would lead to the expectation that creative ideas should be Poisson distributed and randomly scattered over the course of the career.

However, this argument overlooks a very significant fact: The kinds of problems that are most amenable to algorithmic or strong methods tend not to fall in the class of problems that would be considered highly creative. On the contrary, such techniques are most applicable to routine problem solving within a particular discipline. In contrast, scientific creativity is associated with problems in which one or more of the components of the problem space are “ill-defined” (Klahr & Simon, 1999). Thus, the initial state may be poorly specified, the operators may be multiple and diverse, the constraints may be vague and unstable, and the goal states may be ambiguous—and any combination of these possibilities. In these instances, scientists must rely on an assortment of heuristics, or “weak methods” (Klahr, 2000). Examples include the use of analogy, hill climbing, means-end analysis, and just plain trial and error. In general, the more ill defined is the problem space, the

larger the number of potential heuristics and the lower the probability of solution by any given heuristic (Simonton, 1999b). Indeed, just as one may speak of steep and flat associative gradients, so may one think in terms of steep and flat hierarchies with respect to available heuristic methods. The problems that are most likely to elicit solutions deemed creative are those with flat hierarchies in which a large number of heuristic techniques are available, each with approximately equal a priori odds of success.

Earlier I observed that creative scientists tend to work on several projects simultaneously—some are rather routine, others far more risky, even impossible. In terms of the foregoing discussion, the merits of this research strategy should be immediately apparent. Although the routine problems can be readily solved using algorithmic methods, these problems are far less likely to be judged creative by colleagues. The solution of high-risk problems, in contrast, contribute far more to the scientist's reputation as a creative contributor to the field, yet the solutions are more iffy because of their reliance on weak, heuristic strategies. By pursuing a mixture of problems having diverse probabilities of successful solution, a scientist helps ensure that at least something of value will emerge. At the same time, it seems likely that really good ideas will be relatively rare and that those high-impact ideas will be randomly distributed across the career. In short, the behavioral consequences of these problem-solving strategies would be the stochastic phenomena documented earlier.

This expectation is augmented even more when other experimental research is carefully considered. This becomes quite evident when I discuss certain key findings regarding (a) insight problems, (b) creative production, (c) complex inferences, (d) computer problem solving, and (e) group creativity.

Insight Problems

Not all problem-solving tasks are the same. Besides such distinctions as that between well-defined and ill-defined problems, problems can be distinguished according to whether they are "reasonable" or "unreasonable" (Perkins, 2000). Reasonable problems are of the kind that can be solved in a step-by-step manner. A crossword puzzle is of this nature. Given a sufficient vocabulary, the empty spaces can be filled in one by one. Unreasonable problems, in contrast, cannot be treated in this way because the task contains some trick or catch that must be understood before the solution can obtain. This feature derails any step-by-step process that proceeds without the realization that things aren't what they seem. Hence, successful problem solving in these cases requires that the person acquire an insight into the nature of the trick. Riddles provide classic instances of such insight problems.

Experimental research on insight problems is very old, dating back to the early Gestalt psychologists (Köhler, 1925). This tradition was then continued by many cognitive psychologists interested in problem-solving processes that more closely approximate genuine creativity. As a result, a rather large literature has grown on this subject (Sternberg & Davidson, 1995). Of special interest here is the research on the incubation period of the creative process (Wallas, 1926). This phenomenon occurs whenever the problem cannot be immediately solved and thus ends up being placed in the back of the mind or put on the back burner while the scientist turns to other problems that seem more amenable to solution. Although cognitive psychologists do not completely agree on what exactly is

happening during this phase of the creative process, one fact is undeniable: During the incubation period the scientist's mind undergoes exposure to a host of experiences that may or may not be relevant to a successful solution. These experiences are of two kinds: (a) external events in the outside world, including everyday events that normally would not be considered germane; and (b) internal mental events involving thoughts about other projects on which the scientist is actively working. This influx of internal and external experiences can then prime divergent associative pathways (via *spreading activation*; see Martindale, 1995). One of these pathways may eventually lead to the key concept that solves the problem that has been put on hold but not forgotten.

Hence, the incubation period can be said to entail an *opportunistic assimilation* of serendipitous experiences that sets the associative process on a more fruitful track (Seifert, Meyer, Davidson, Patalano, & Yaniv, 1995). Moreover, the length of the incubation period will be largely determined by pure luck (Perkins, 2000; Simonton, 1988b). The experiential influx is not directed by the problem itself but rather by other problems and even general life experiences. Sometimes the precipitating input arrives within hours, and other times it may take weeks, even years, without the scientist being able to either control or anticipate the course of events (Simonton, 1995). Indeed, should the scientist attempt to take deliberate conscious control over the incubation process, the immediate consequence would be a restriction in the necessary openness to "irrelevant" influences, thereby undermining its effectiveness. It is for this reason that insight problems are much more easily solved intuitively than by applying explicitly conscious logical inference (Schooler & Melcher, 1995). Among the diverse inputs that are available to intuitive processing but denied to conscious processing is tacit knowledge that might lead the associative process toward more productive directions (Reber, 1993).

It should be clear that the incubation process operates in such a way as to render scientific creativity highly stochastic in its behavioral manifestations. Certainly the operation is consistent with the empirical finding that creative ideas tend to be randomly distributed across the career course.

Creative Production

One major objection that can be raised against experimental research on insight problems concerns the nature of the task. The participants in these experiments are presented with problems with known solutions. Genuine creative behavior seldom works this way, in science or otherwise. Instead, creativity is applied to open-ended problems in which the answer is unknown. Indeed, nobody may know for sure whether an answer even exists. Therefore, to understand better how creativity works in real-life settings, inquiry must turn to the generation of creative products. That is, participants can be asked to produce original ideas under various experimental conditions, and the creativity of those ideas can then be evaluated (Amabile, 1982). Investigations using this approach have demonstrated that creativity is enhanced when research participants are exposed to quasi-random or incongruous stimuli (Finke, Ward, & Smith, 1992; Proctor, 1993; Rothenberg, 1986; Sobel & Rothenberg, 1980).

The experiments conducted by Finke et al. (1992) are most relevant to the present analysis. The participants were provided shapes or forms—lines, circles, triangles, letters, spheres, cubes,

rings, hooks, and so forth—from which they were instructed to create objects with recognizable functions (e.g., furniture, appliances, tools and utensils, weapons, or toys). Judges then independently evaluated the creativity of the resulting inventions. In some experimental conditions the participants could select the shapes for the imaginative constructions, whereas in other conditions the participants were simply given a random selection of forms. In yet another manipulation the participants themselves could choose the category of object they had to invent or the category would be selected randomly by the experimenter. The investigators showed that participants arrived at the most innovative solutions when both the object parts that they had to work with and the category of object they had to invent were randomly selected from the larger pool of possibilities. These experiments can be viewed as simulations of serendipitous events that stimulate the creativity of scientists who are sufficiently open to novel experiences.

Complex Inferences

As noted at the outset of this article, Simon (1973) held that even the most momentous scientific discoveries involved the straightforward application of simple logic. For instance, he claimed that Mendeléeve's origination of the Periodic Law of the Elements required nothing more advanced than what is "required to handle patterned letter sequences" (p. 479). The experimental research on scientific reasoning suggests that the process is far more complex and tenuous than Simon's claim implies (Faust, 1984). The human intellect is not well equipped to make inferences from information that is highly ambiguous, inconsistent, probabilistic, multidimensional, nonadditive, or nonlinear (Fiske & Taylor, 1991; Kahneman, Slovic, & Tversky, 1982). Basic limits in human information processing and certain biases in inference combine to undermine inductive and deductive reasoning in even highly intelligent scientists (Mahoney, 1976; Mitroff, 1974). A discovery often becomes logically obvious only after it has been made—an impression that is itself the product of the hindsight bias (Fischhoff, 1975).

These experimental results provide an additional empirical basis for understanding the stochastic nature of creative behavior in science. A case in point is the equal-odds rule, in both longitudinal and cross-sectional forms. If it was very easy to acquire inferential tools that lead so inevitably to breakthrough discoveries, then scientists should improve their hit rates over the course of their careers. In addition, those who have been the most prolific should have acquired more inferential expertise than those who have been much less so and thereby should have higher hit rates. Yet such improvement does not appear to be possible. A scientist might infer some principle that enabled him or her to solve some problem, but that inference may be wrong, so that the scientist discovers that the same approach falters on the next problem. The Einstein who conceived the unsuccessful unified field theory was the same Einstein who had earlier created the high-impact theories of general and special relativity. Aggravating matters all the more is the inconsistent feedback scientists receive from peer review. Given the low reliability and predictive validity of the evaluations allotted to submitted manuscripts and grant proposals, judging which techniques work and which do not is no easy task. As a result, the output of high-impact ideas must necessarily be a hit or

miss affair, yielding the stochastic pattern of creative productivity across and within careers.

Computer Problem Solving

I have just shown that the experimental literature on insight problems, creative production, and complex inferences all support the thesis that creative behavior in science must be fundamentally stochastic. Yet this conclusion might seem to be contradicted by some computer simulations of the creative process. In particular, investigators have written programs that claim to make breakthrough discoveries using inherently logical, step-by-step procedures (Langley et al., 1987; Shrager, & Langley, 1990). For example, one of these discovery programs, BACON, which implements inductive logic, was credited with rediscovering Kepler's Third Law of planetary motion, Black's Law of temperature equilibrium, Ohm's Law of current and resistance, Prout's hypothesis of atomic structure, the Gay-Lussac Law of gaseous reaction, the Dulong-Petit Law of atomic heats, and the derivation of atomic weights by Avogadro and Cannizzaro (Bradshaw, Langley, & Simon, 1983).

Such achievements look very impressive and seem to make a strong case in favor of the thesis that scientific creativity has a deterministic logic quite contrary to the stochastic combinatorial process outlined here. Even so, these programs fail to stand up to detailed scrutiny (Simonton, 1999b; Sternberg 1989; Tweney, 1990). These discovery programs are actually *rediscovery* programs specifically designed to discover what has already been discovered. Hence, they are not creative in any meaningful sense of the word. In contrast, computer models that come closest to duplicating human creativity all incorporate some kind of stochastic mechanism (Boden, 1991, 1995; Johnson-Laird, 1993). For example, the problem-solving programs known as genetic algorithms and genetic programming depend on the introduction of a random combinatorial process (Goldberg, 1989; Koza, 1992, 1994). These programs have not only made rediscoveries, such as Kepler's Third Law, but have also generated truly creative solutions to real problems. To be sure, computers must advance much farther before they can be said to simulate all essential features of scientific creativity. For instance, no computer program simulates the cross-talk between several projects being carried on simultaneously or the opportunistic assimilation of serendipitous events from the external world (Simonton, 1999b; Tweney, 1990). Yet if ever such programs emerge, it is most likely that the resulting creative behavior will be more unpredictable rather than less so (see also Martindale, 1995).

Group Creativity

In one respect all of the foregoing studies overlook a major feature of modern scientific creativity, namely, that it now largely takes place in collaborative groups, particularly in research laboratories located in either academic or industrial settings. For example, a typical scientific laboratory at a research university might consist of the principal investigator, various coinvestigators, post-doctoral and doctoral students, laboratory technicians, and perhaps even advanced undergraduates. Therefore, to understand more fully the creative process in science requires a better appreciation for how creative ideas might emerge from interactions among

group members. Unfortunately, most laboratory experiments on this subject use unrepresentative participants (viz., college students) and unrepresentative problem-solving tasks (e.g., ones that require no division of labor or special expertise), thereby undermining their generalizability to real-world problem-solving groups. Even so, two experimental findings are of special value in the current context:

1. Inquiries into group brainstorming has divulged how idea sharing among group members can result in cognitive stimulation of each member (Dugosh, Paulus, Roland, & Yang, 2000). In other words, by exchanging ideas each individual in the group can be stimulated in an unexpected manner, thereby providing another source of external serendipitous influence on the person's train of thought.
2. Group problem solving appears to be enhanced when the members are exposed to a minority that departs from the majority opinion (Nemeth & Wachtler, 1983). Specifically, members so exposed would discover new solutions to the problems that were offered by neither the majority nor the minority and yet that would tend to be correct rather than incorrect.

These experimental findings taken together suggest that a research team whose membership is quite heterogeneous with respect to expertise, experience, and status may prove more creative than one that is far more homogeneous. Essentially, information exchange among separate members functions in a manner similar to what happens within a single scientist who is simultaneously engaged on several distinct projects, so that the work on one project can lead to serendipitous priming of other products undergoing the incubation phase. This expectation has been confirmed in empirical research. For example, a study of 1,222 research units in six European countries found that the rated effectiveness and objective productivity was positively correlated with diversity with respect to disciplines, specialties, funding sources, professional functions, and projects (Andrews, 1979; see also Dunbar, 1995). Yet it is evident that the consequence of such diversity would be groups that are not only more creative, but also more unpredictable in their creative output. Major discoveries would be randomly distributed across time at the group level in the same way that they are at the individual level.

Toward the Integration of the Three Perspectives

I have tried to demonstrate that the stochastic structure of scientific creativity can be explicated in terms of both in vitro traditions. On the one hand, the correlational (psychometric) literature has divulged cognitive and dispositional characteristics of the creative person that would seem to support the highly unpredictable behavior seen in scientific creativity. These personal traits include the capacity for generating unusual associations and the tendency to be open to novel or surprising experiences. On the other hand, the experimental (laboratory) literature, although seemingly antithetical to the stochastic premise, can also be construed to be supportive. In particular, the accumulated research on insight problems, creative production, complex inferences, com-

puter problem solving, and group creativity all suggests that the creative process is far less logical and deterministic than often claimed. When these two research traditions are combined with the behavioral patterns of real-life scientific careers and communities, the stochastic nature of creative behavior becomes all the more unmistakable.

Given these conceptual linkages, two additional questions must be addressed to obtain a comprehensive psychological understanding of scientific creativity. First, how might the person and process perspectives be effectively unified? Second, why must the product perspective be integrated with the person and process perspectives to form any complete account?

Integrating the Person and Process Perspectives

If cross-talk between separate projects can contribute to insightful problem solving, perhaps the field can benefit from an analogous exchange between cognitive psychologists who study the creative process and differential psychologists who study the creative person. Admittedly, some psychologists in the experimental tradition may resist this suggested integration—a suggestion that, after all, comes from a psychologist more strongly affiliated with the correlational tradition. In the first place, many cognitive psychologists insist that individual-difference variables have little explanatory utility, any contrasts in scientific creativity reflecting differences in domain knowledge and problem-solving expertise (Hayes, 1989b; Klahr & Simon, 1999; Weisberg, 1992). Furthermore, these same investigators tend to be convinced that scientific creativity has a definite logic that can be teased out in laboratory experiments and computer simulations (e.g., Klahr, 2000; Shrager & Langley, 1990). To argue otherwise is often interpreted as an attempt to mystify creativity by conjuring up “special processes” (Perkins, 2000; Weisberg, 1992). However, before the suggested integration is rejected out of hand, I think that all psychologists should weigh the following two considerations.

First, a strong *prima facie* case can be made that certain individual-difference variables must have direct effects on the cognitive processes involved in creative problem solving. For instance, if successful resolution of the incubation period often involves the opportunistic assimilation of stimuli that prime more fruitful associative pathways, then this process would certainly be affected by individual differences in negative priming and latent inhibition. Those who less effectively filter out seemingly irrelevant priming stimuli will necessarily survey a wider range of associative paths. Likewise, if the priming process has such a potent role in insightful solutions, it seems probable that individual differences in associative richness would strongly moderate the impact of priming stimuli by determining the extent of the spreading activation that results. The richer the associative base, the more pervasive and less constrained the activation. In the main, it is difficult to conceive of a basic process involved in scientific creativity that would not be either enhanced or inhibited by at least one individual-difference variable that has been empirically associated with actual creative behavior. This statement holds even for more purely dispositional variables, such as openness to experience and tolerance of ambiguity, two traits that would certainly increase the unpredictability of the associative process. I must emphasize that this argument is not tantamount to the assertion that scientific discoveries require any special processes, even less mys-

terious ones. A sports analogy may make this clear: Tall basketball players may average more points per game than do short players, but that does not mean that the former are taking advantage of some special magic tricks unavailable to the latter. The basic sensory and motor mechanics of shooting may be largely the same, with only adjustments to the kinds of defense that a taller player is more likely to encounter on the court.

Second, to claim that creative behavior is stochastic in structure is not equivalent to maintaining that logical processes have no place whatsoever in scientific productivity. For one thing, scientific contributions display considerable variation in the amount of creativity they require—from journal articles that report unexpected breakthroughs to scholarly research summaries such as those that dominate the chapters in edited volumes. The less creativity demanded by a given publication, the more logical processes prevail over those that are purely stochastic. Even in the case of contributions that presuppose considerable creativity, it is advisable to distinguish between the “context of discovery” and the “context of justification” (Reichenbach, 1938; see also Popper, 1959). To comprehend discovery requires a grasp of the psychological processes responsible for that event, whereas to understand justification demands an appreciation of the arguments by which the scientific truth of that discovery is established. Furthermore, the discovery process must occur prior to the justification process, and to a large extent the two processes must be independent and distinct. This primacy, independence, and distinctiveness is evident in Albert Einstein’s description of his own creative process: “Taken from a psychological viewpoint . . . combinatory play seems to be the essential feature in productive thought—before there is any connection with logical construction in words or other kinds of signs which can be communicated to others” (as cited in Hadamard, 1945, p. 142). Consequently, “conventional words or other signs have to be sought for laboriously only in a secondary stage, when the mentioned associative play is sufficiently established and can be reproduced at will” (as cited in Hadamard, 1945, p. 143).

Even when attention is restricted to the context of discovery, the stochastic model of scientific creativity cannot be said to exclude logic. On the contrary, the claim is limited to the assertion that scientific behavior at the macro level fails to be logically determined. Although logic has a role to play at the micro level, inductive and deductive reasoning alone do not suffice to reproduce the psychological phenomenon of creative behavior. Instead, logical processes are coupled with other cognitive processes that, when aggregated together, yield the stochastic outcomes described in this article. In fact, it might even be argued that the same stochastic behavior would appear at the macro level even if the underlying processes at the micro level were entirely logical in structure. To make this argument, I can draw an analogy with the kinetic theory of gases that Poincaré (1921) used as a simile for the creative process. At the molecular level, a gas can be conceived as a multitude of perfectly elastic but extremely small billiard balls that bounce off each other according to purely deterministic mechanical laws. Moreover, each of these collisions is perfectly reversible in time. Yet the sum total of these collisions at the molar level is the nondeterministic and irreversible behavior described by statistical mechanics and the second law of thermodynamics (e.g., Brownian motion). In a similar fashion, many undeniably logical

acts can produce aggregate behavior that appears devoid of any logic.

Incorporating the Product Perspective

If both person and process perspectives are truly compatible with the product perspective, and if the person and process perspectives prove compatible with each other, then one might ask whether it is really necessary to include the product perspective in a integrative scientific explanation. After all, I have argued that the stochastic processes implicated in scientific careers and communities can be explicated in terms of the research findings of the correlational and experimental traditions. Even so, there are at least three reasons why the product perspective must be joined with the other two if we seek a complete analysis of the phenomenon.

First, although the stochastic model of scientific creativity can be derived from the correlational and experimental research traditions, it not obvious that either of these traditions would have drawn this important implication. On the one hand, the person perspective has close ties with mysterious notions of geniuses as having access to special processes or abilities that enable them to make discoveries in some infallible flash of intuition or insight. On the other hand, the process perspective has an intimate connection with the rationalist belief that scientific creativity represents nothing more than the application of a precise and logical method. In either case, it is not self-evident that either of these traditions would spontaneously generate such probabilistic derivations as the equal-odds rule or the Poisson model of multiple discovery. Indeed, neither tradition did so. The cause of this oversight is pretty clear: Investigators in both research traditions cannot see the forest for the trees. The stochastic nature of scientific creativity does not emerge as an indubitable reality until the inquiry turns to how products are distributed across and within careers as well as within communities of scientists. Exclusive focus on the personality traits of creators or the cognitive processes behind creativity causes psychologists to overlook a critical aspect of a phenomenon that transcends both traits and processes.

Second, a complete integration of the three alternative perspectives on scientific creativity may allow the phenomenon to become part of a larger, more comprehensive psychological synthesis. Currently many psychologists have proposed that evolutionary psychology enjoys great promise of providing an integrative framework for all or most of the discipline (Barkow, Cosmides, & Tooby, 1992; Crawford & Krebs, 1998). Furthermore, a stochastic combinatorial model of scientific discovery can be subsumed under a Darwinian theory of creativity (Campbell, 1960; Eysenck, 1995; Findlay & Lumsden, 1988; Simonton, 1999b). Indeed, the genetic algorithms that have proved so effective in problem solving are explicitly connected to evolutionary theory (Goldberg, 1989; Martindale, 1995). If conceptual unity is an important feature of mature science, then this unification would directly contribute to the advancement of psychology as a science.

Third and last, even if an evolutionary integration is viewed as impossible or undesirable, the unification of the person, process, and product perspectives should still enhance psychological understanding of what must be considered an intrinsically important phenomenon: scientific creativity. This phenomenon is important to all sciences, including psychology (Simonton, 1988b, 2002).

Yet it should be evident that this phenomenon has three essential components: (a) the products that contain the creative ideas, (b) the persons who conceived those ideas, and (c) the processes those persons used to do so (Simonton, 2000). Accordingly, a complete comprehension of scientific creativity demands that all three components be subject to scientific investigation.

The importance of the proposed integration may be illustrated by elaborating on the expression "not to see the forest for the trees." The person- and process-oriented psychologists are comparable to scientists who investigate singular trees. The person-oriented psychologists are botanists who focus on how trees vary in a diversity of morphological traits, such as how firs differ from pines with respect to leaves, cones, branches, bark, trunk, and roots. The process-oriented psychologists are plant physiologists who analyze trees in terms of basic mechanisms, such as the role of osmotic and capillary processes in the extraction and conveyance of water and nutrients, the genetic processes behind reproduction, and the photosynthetic processes by which trees support their metabolism. Although both botanists and plant physiologists provide us with everything we may want to know about individual trees, they cannot provide the whole picture. After all, a significant characteristic of most trees is that they tend to be part of ecological systems, especially forests, with distinct properties that cannot be reduced to either botany or plant physiology. These attributes include the distribution of different tree species relative to geography, rainfall, temperature, soil, flora and fauna, and other conditions. By the same token, product-oriented psychologists scrutinize how discoveries are distributed across individual scientists (as affected by the characteristics of research programs and peer review) and scientific communities (as affected by the attributes of the domain and disciplinary communication). As a consequence, they are studying the forest, not the trees. Yet if you compile everything that the botanist, plant physiologist, and ecologist can tell you about their respective findings, you obtain a complete knowledge of trees, both as singular plants and as the collectives known as forests. In the same way, the *in vivo* behavioral inquiries into scientific careers and communities must be integrated with *in vitro* studies from the two disciplines of psychology, one concentrating on the person who creates and the other on the process of creation. To do less will leave psychology with only a fragmentary and misleading perspective on scientific creativity.

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